

Daily Risk-Asset Trade Signals



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7 min read · Apr 22, 2022



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Here's how I used Kaggle's schedule feature to build a daily cryptocurrency trade signals webpage.

See the program here: [Daily Binance Trade Signals](#)





Kaggle's trigger feature. CCXT (<https://github.com/ccxt/ccxt>) is an exchange API wrapper that supports a stupendous list of brokerages and is incredibly easy to use. Pandas Technical Analysis Library (<https://github.com/twopirllc/pandas-ta>) serves as our pool of indicators & forecasting algorithms. And the schedule feature within Kaggle's notebook kernel enables for daily, weekly, or monthly executions of code will be our vehicle for daily automation.

Date	Close	Ichimoku 9-day Forecast	Ichimoku 26-day Forecast	Choppiness (%)	RSI	Archer MA Trends	EMA Crossover	Buy	Sell	Tick	Deno
2022-04-21	0.000009	0.000012	0.000024	62.38	35.095811	True	False	False	False	XRP	BTC
2022-04-21	20.237800	31.309850	37.399100	39.67	35.339421	False	False	False	False	WAVES	USD
2022-04-21	2.106600	2.410975	3.010000	55.37	35.658788	False	False	False	False	NANO	USD
2022-04-21	18.991900	21.473850	23.064350	43.69	36.077216	False	False	False	False	HNT	USD
2022-04-21	0.000339	0.000356	0.000362	54.23	38.548572	False	False	False	False	LINK	BTC
2022-04-21	0.118000	0.129522	0.142540	53.90	38.920330	False	False	False	False	ONE	USDT
2022-04-21	46.160000	52.930000	59.290000	48.34	38.957552	False	False	False	False	AXS	USD
2022-04-21	0.022700	0.023950	0.025450	43.64	39.619105	False	False	False	False	AMP	USD
2022-04-21	0.000023	0.000024	0.000024	48.10	39.659244	False	False	False	False	ADA	BTC
2022-04-21	2.105300	2.249825	2.442700	51.07	40.004355	False	False	False	False	MANA	USD

Install & Import Packages

```
# login to binance
from IPython.display import clear_output

!pip install ccxt
!pip install schedule
!pip install pandas_ta

import ccxt,schedule,warnings,time,ast
warnings.filterwarnings('ignore')
import matplotlib.pyplot as plt
from dateutil.tz import tzlocal
```

```
from rich import print, pretty
from datetime import datetime
from random import randint
from random import seed
import pandas_ta as ta
import pandas as pd
import numpy as np
pretty.install()
```

Login To Binance.us

The following is respective of ccxt's login protocol for binance.us.

(<https://docs.ccxt.com/en/latest/exchange-markets.html>)

```
# Instructions on retrieving your own API key pair:
# https://youtu.be/Q4Lt1Kulk0Q

ccxt.binanceus({ 'options':{ 'adjustForTimeDifference':True}})
exchange = ccxt.binanceus({
    "apiKey": Binance_key,
    "secret": Binance_secret,
    'enableRateLimit': True})
```

Calculate Indicators

What I find most valuable about this model is that I can easily customize & fine-tune my buy/sell trade signal parameters. I decided to use the following technical indicators & forecasting algorithms in order to “recommend” (i.e. discern against) going long or going short: Kalman Filter Forecast Model, 14-Day Lower/Upper Bollinger Bands, 9 & 26-Day Senkou Span Ichimoku, Archer Moving Averages Trends (AMAT), & Relative Strength Index (RSI). Although the 14/91/125-Day Exponential Moving Averages (EMA) & Choppiness Index were calculated, they are not currently used as decision parameters for the output below

```
# indicators + trade signals
```

```

def calculate_indicator(symbol):

    bars = exchange.fetch_ohlcv(symbol, timeframe='1d', limit=300)
    df = pd.DataFrame(bars[:-1], columns=['timestamp', 'open',
    'high', 'low', 'close', 'volume'])
    df['timestamp'] = pd.to_datetime(df['timestamp'],
    unit='ms').dt.tz_localize(None)

    close = df['close'][len(df)-1]
    low = df['low'][len(df)-1]

    # Construct a Kalman filter
    kf = KalmanFilter(transition_matrices = [1],      # The value for
    At. It is a random walk so is set to 1.0
                    observation_matrices = [1],      # The value for
    Ht.
                    initial_state_mean = 0,          # Any initial
    value. It will converge to the true state value.
                    initial_state_covariance = 1,    # Sigma value for
    the Qt in Equation (1) the Gaussian distribution
                    observation_covariance=1,        # Sigma value for
    the Rt in Equation (2) the Gaussian distribution
                    transition_covariance=.01)       # A small
    turbulence in the random walk parameter 1.0
    # Get the Kalman smoothing
    state_means, _ = kf.filter(df['close'].values)
    # Call it kf_mean
    df['kf_mean'] = np.array(state_means)
    kalman = df.kf_mean[len(df)-1]
    aboveKalman = low > kalman

    # exponential moving averages
    ema_14 = df.ta.ema(14, append=True)[-1:].reset_index(drop=True)
[0]
    ema_91 = df.ta.ema(91, append=True)[-1:].reset_index(drop=True)
[0]
    ema_125 = df.ta.ema(125, append=True)[-1:].reset_index(drop=True)
[0]
    ema_crossover = ema_14 > ema_91

    # lower/upper 14-day bollinger bands for mean reversion
    bbl_14 = df.ta.bbands(length=14, append=True)
[['BBL_14_2.0']].tail(1).values[0][0]
    bbu_14 = df.ta.bbands(length=14, append=True)
[['BBU_14_2.0']].tail(1).values[0][0]
    bband_buy = close < bbl_14
    bband_sell = close > bbu_14

    # ichimoku 9 & 26-day forecasts
    # https://technical-analysis-library-in-
    python.readthedocs.io/en/latest/ta.html#ta.trend.IchimokuIndicator
    isa_9 = df.ta.ichimoku()[1]['ISA_9'].tail(1).values[0] #
    help(ta.ichimoku)
    isb_26 = df.ta.ichimoku()[1]['ISB_26'].tail(1).values[0]

```

```

# archer ma
# https://github.com/twopirllc/pandas-ta#general
amat = (df.ta.amat()['AMATe_LR_8_21_2'].tail(1).values[0] == 1)

# rsi
rsi = df.ta.rsi()[len(df)-1]
rsi_buy = rsi < 30
rsi_sell = rsi > 70

# choppy
# https://github.com/twopirllc/pandas-ta#trend-18
try:
    chop = "{:.2f}".format(df.ta.chop()[len(df.ta.chop())-1])
except RunTimeWarning:
    chop = 0

# ~~~~~ #
# signal
#buy = (close < isa_9) & (close < isb_26) & amat & rsi_buy &
bband_buy & aboveKalman
buy = amat & ema_crossover & aboveKalman

#sell = (close > isa_9) & (close > isb_26) & ~amat & rsi_sell &
bband_sell & ~aboveKalman
sell = ~amat & ~ema_crossover & ~aboveKalman

# ~~~~~ #

return df, symbol, close, isa_9, isb_26, chop, rsi, amat,
ema_crossover, buy, sell, aboveKalman

```

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```

        'Ichimoku 26-day Forecast':output[4],
        'Choppiness (%)':output[5],
        'RSI':output[6],
        'Archer MA Trends':output[7],
        'EMA Crossover':output[8],
        'Buy':output[9],
        'Sell':output[10]
    })

    except:
        pass

# frame & clean
results = pd.DataFrame(results)
results['Tick'] = [v.split('/')[0] for i,v in results.Ticker.items()]
results['Deno'] = [v.split('/')[1] for i,v in results.Ticker.items()]

# set pandas display to respect number of symbols in results
size = len(results)
pd.set_option('display.max_rows', size)

```

Top & Bottom 10s

```

# top 10 recommended buys
top10 = results.sort_values(['Buy','EMA Crossover','Archer MA
Trends'],ascending=False).sort_values(['RSI'],ascending=True).drop_du
plicates('Tick').head(10).reset_index(drop=True)

# top 10 recommended sells
bottom10 = results.sort_values(['Sell','Archer MA
Trends','RSI'],ascending=False).drop_duplicates('Tick').head(10).rese
t_index(drop=True)
print("Prepped.")

```

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If any of the results in our top 10 are recommended Buys, I'd want to see

those first. But if every result came out False, then there are no recommended long positions.

```
if any(results.Buy == True):
    print("These are recommended long positions, then sorted by
crossover, trend, & rsi values.")
if all(results.Buy == False):
    print("No recommended long positions. Showing results sorted by
crossover, trend, & rsi values.")
```

I do the same concept for Sells:

```
if any(results.Sell == True):
    print("These are recommended short positions, then sorted by
crossover, trend, & rsi values.")
if all(results.Sell == False):
    print("No recommended short positions. Showing results sorted by
trend & rsi values.")
```

Recommendations

```
# plots for top3
[plot(symbol) for i,symbol in top3.Ticker.items()]
```

Top 3 Buys

```
# plots for bottom3  
[plot(symbol) for i,symbol in bottom3.Ticker.items()]
```

Bottom 3 Sells

Scheduler

Located at the far right of Kaggle's notebook panel > Settings, you'll find a trigger feature that allows for daily, weekly, or monthly executions of the kernel. This provides me with all the data I need in order to get a pulse on current market trends.

In all honesty, I'm more excited about the fact that I have something to look at and revel everyday. Only to find fallacies, ways to be more efficient, or simple changes need to be made so as to give me something fun to do on some idle Saturday morning.

Key Takeaways

With that said, it's important to note (even for myself) that the signals provided will be completely dependent on the conditions I set within the `calculate_indicator()` protocol. This gives me frequent opportunities to better tune my forecasting models, clean out some inefficiencies, or add more indicators as I continue to learn about the fluctuations of the crypto market.

On-going & Future Efforts

It would be great to use Kaggle's scheduler feature in order to place actual trades. Implementing more sophisticated indicators, robust forecasts, safe limit orders, & stop-loss thresholds might someday serve as a comfortable vehicle for automated trading. This protocol can be written to run against other exchanges, of course, like TD Ameritrade & Robinhood which both have open source APIs. This project will be on-going and I expect to continue

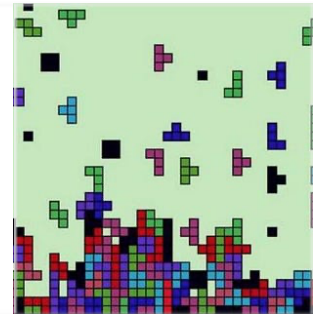
tuning the notebook's recommendations. In other words, it will never be perfect.

<https://linkedin.com/in/dontadaya>

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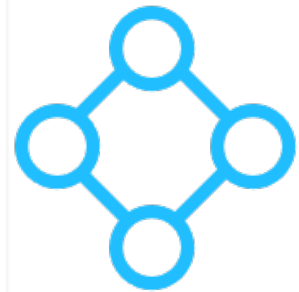
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